

DATA STRUCTURES - IT23302 HW ASSIGNMENT SET - 3

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1. Write a C program to store n elements in an array using dynamic memory allocation and find the largest number

```
2. #include<stdio.h>
3. #include<stdlib.h>
4. int main()
5. {
6.     int *arr;
7.     int max=-1;
8.     int n;
9.     printf("Size of Array : ");
10.    scanf("%d",&n);
11.    arr=(int*)calloc(n,sizeof(int));
12.    printf("Enter array Elements : ");
13.    for(int i=0;i<n;i++)
14.    {
15.        scanf("%d",&arr[i]);
16.        if(arr[i]>max) max=arr[i];
17.    }
18.    printf("Largest element : %d",max);
19.    return 0;
20.
21. }
```

OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./1

Size of Array : 3

Enter array Elements : 1

2

4

Largest element : 4

2. Print an array in reverse order using pointers.

```
#include<stdio.h>
int main()
```

```

{
    int arr[100];
    int *aptr=arr;
    int max=-1;
    int n;
    printf("Size of Array : ");
    scanf("%d",&n);
    printf("Enter array Elements : ");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("Array in Reverse Order : ");
    for(int i=n-1;i>=0;i--)
    {
        printf("%d ",*(aptr+i));
    }
    return 0;
}

```

OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./2

Size of Array : 3

Enter array Elements : 1

1

3

Array in Reverse Order3 1 1

3. Write a C program to store information of students using structure. Sort them by Name.

```

#include<stdio.h>
#include<string.h>
struct Students
{
    long int roll;
    char name[100];
};
int main()
{
    int n;
    struct Students temp;

```

```

printf("Enter the number of students");
scanf("%d",&n);
struct Students s[n];
printf("Enter Student Details (Roll Number and Name)");
for(int i=0;i<n;i++)
{
    scanf("%ld",&s[i].roll);
    scanf("%s",s[i].name);
}
for(int i=0;i<n;i++)
{
    for(int j=0;j<n-i-1;j++)
    {
        if(strcmp(s[j].name,s[j+1].name)>0)
        {
            temp=s[j];
            s[j]=s[j+1];
            s[j+1]=temp;
        }
    }
}
printf("After sorting ,Student Details \n");
printf("Enter Student Details (Roll Number and Name)");
for(int i=0;i<n;i++)
{
    printf("%ld\n",s[i].roll);
    printf("%s\n",s[i].name);
}
return 0;
}

```

OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./3

Enter the number of students2

Enter Student Details (Roll Number and Name)2025115120

RAHUL

2025103029

RJ

After sorting ,Student Details

Enter Student Details (Roll Number and Name)2025115120

RAHUL

2025103029

RJ

4. Write a C program to add n matrices given in each set using structure.

```
#include<stdio.h>
struct distance{
    int feet;
    int inch;
};
int main()
{
    int n;
    int feet=0,inch=0;
    printf("Enter the number of distances");
    scanf("%d",&n);
    struct distance s[n];
    printf("Enter distance Details (Feet and Inches)");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&s[i].feet);
        scanf("%d",&s[i].inch);
        feet+=s[i].feet;
        inch+=s[i].inch;
    }
    feet+=inch/12;
    inch=inch%12;
    printf("Total distance = %d feet %d inches",feet,inch);
    return 0;
}
```

OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./4

Enter the number of distances2

Enter distance Details (Feet and Inches12

14

13

16

Total distance = 27feet 6 inches

5. Write a C program to add two complex numbers by passing structure to a function.

```
#include<stdio.h>
struct Complex
```

```

{
    int real;
    int imag;
};
void add(struct Complex c1, struct Complex c2, struct Complex *ans);
int main()
{
    struct Complex c1,c2,ans;
    printf("Enter Real Part of 1st number : ");
    scanf("%d",&c1.real);
    printf("Enter Imaginary Part of 1st number : ");
    scanf("%d",&c1.imag);
    printf("Enter Real Part of 2nd number : ");
    scanf("%d",&c2.real);
    printf("Enter Imaginary Part of 2nd number : ");
    scanf("%d",&c2.imag);
    add(c1,c2,&ans);
    printf("Added Complex number : %d + %di",ans.real,ans.imag);
}
void add(struct Complex c1,struct Complex c2,struct Complex *ans)
{
    ans->real=c1.real+c2.real;
    ans->imag=c1.imag+c2.imag;
}

```

OUTPUT

```

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./5
Enter Real Part of 1st number : 10
Enter Imaginary Part of 1st number : 20
Enter Real Part of 2nd number : 15
Enter Imaginary Part of 2nd number : 25
Added Complex number : 25 + 45i

```

6. Write a C program to maintain library management system using structure.

```

#include<stdio.h>
#include<string.h>
struct Product
{

```

```

    long int productid;
    char name[100];
    float price;

};

int main()
{
    int n;
    printf("Enter the number of Products");
    scanf("%d",&n);
    struct Product p[n];
    printf("Enter Product Details (ID ,Name and price)");
    for(int i=0;i<n;i++)
    {
        printf("Enter Product ID : ");
        scanf("%ld",&p[i].productid);
        printf("Enter Product Name : ");
        scanf("%s",p[i].name);
        printf("Enter Product Price : ");
        scanf("%f",&p[i].price);

    }
    printf("Displaying the details\n");
    for(int i=0;i<n;i++)
    {
        printf("Product ID : ");
        printf("%ld\n",p[i].productid);
        printf("Product Name : ");
        printf("%s\n",p[i].name);
        printf("Product Price : ");
        printf("%f\n",p[i].price);

    }
    return 0;
}

```

OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./6

Enter the number ofProducts2

Enter Product Details (ID ,Name and price)Enter Product ID : 1

Enter Product Name : eARPODS

Enter Product Price: 2000
Enter Product ID : 2
Enter Product Name : TV
Enter Product Price: 20000
Displaying the details
Product ID : 1
Product Name : eARPODS
Product Price : 2000.000000
Product ID : 2
Product Name : TV
Product Price : 20000.000000

□7. Write a C program for decimal to binary, octal and hexadecimal conversion.

```
#include<stdio.h>

int main()
{
    int dec, temp, rem;

    printf("Enter Decimal Number : ");
    scanf("%d",&dec);

    temp=dec;

    printf("Binary : ");
    if(temp==0)
    {
        printf("0");
    }
    else
    {
        int binary[32];
        int i=0;

        while(temp>0)
        {
            binary[i]=temp%2;
            temp/=2;
            i++;
        }
    }
}
```

```

        for(i=i-1;i>=0;i--)
        {
            printf("%d",binary[i]);
        }
    }

    temp=dec;

    printf("\nOctal : ");
    printf("%o",temp);

    temp=dec;

    printf("\nHexadecimal : ");
    printf("%X",temp);

    return 0;
}

```

□ OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./7

Enter Decimal Number : 25

Binary : 11001

Octal : 31

Hexadecimal : 19

8. Write a C program to calculate difference between two time periods.

```

#include<stdio.h>

struct Time
{
    int hours;
    int minutes;
    int seconds;
};

int main()
{
    struct Time t1,t2,diff;

    printf("Enter First Time (Hours Minutes Seconds) : ");
}

```



```

scanf("%d%d%d",&t1.hours,&t1.minutes,&t1.seconds);

printf("Enter Second Time (Hours Minutes Seconds) : ");
scanf("%d%d%d",&t2.hours,&t2.minutes,&t2.seconds);

if(t1.seconds<t2.seconds)
{
    t1.seconds+=60;
    t1.minutes--;
}

diff.seconds=t1.seconds-t2.seconds;

if(t1.minutes<t2.minutes)
{
    t1.minutes+=60;
    t1.hours--;
}

diff.minutes=t1.minutes-t2.minutes;
diff.hours=t1.hours-t2.hours;

printf("Difference = %d Hours %d Minutes %d Seconds",
        diff.hours,diff.minutes,diff.seconds);

return 0;
}

```

OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./8

Enter First Time (Hours Minutes Seconds) : 10 30 45

Enter Second Time (Hours Minutes Seconds) : 8 20 30

Difference = 2 Hours 10 Minutes 15 Seconds

9. Write a C program to develop an employee management system using structures.

```

#include<stdio.h>

struct Employee
{
    long int id;

```

```

    char name[100];
    float salary;
};

int main()
{
    int n;

    printf("Enter the number of Employees : ");
    scanf("%d",&n);

    struct Employee e[n];

    printf("Enter Employee Details (ID, Name and Salary)\n");

    for(int i=0;i<n;i++)
    {
        printf("\nEnter Employee ID : ");
        scanf("%ld",&e[i].id);

        printf("Enter Employee Name : ");
        scanf("%s",e[i].name);

        printf("Enter Employee Salary : ");
        scanf("%f",&e[i].salary);
    }

    printf("\nEmployee Details\n");

    for(int i=0;i<n;i++)
    {
        printf("\nEmployee ID : %ld",e[i].id);
        printf("\nEmployee Name : %s",e[i].name);
        printf("\nEmployee Salary : %.2f\n",e[i].salary);
    }

    return 0;
}

```

OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./9

Enter the number of Employees : 2

Enter Employee Details (ID, Name and Salary)

Enter Employee ID : 101

Enter Employee Name : RAHUL

Enter Employee Salary : 50000

Enter Employee ID : 102

Enter Employee Name : RJ

Enter Employee Salary : 45000

Employee Details

Employee ID : 101

Employee Name : RAHUL

Employee Salary : 50000.00

Employee ID : 102

Employee Name : RJ

Employee Salary : 45000.00

10. Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

```
#include<stdio.h>

float rectangleArea(float length,float breadth)
{
    return length*breadth;
}

float rectanglePerimeter(float length,float breadth)
{
    return 2*(length+breadth);
}

float triangleArea(float base,float height)
{
    return 0.5*base*height;
}

float trianglePerimeter(float a,float b,float c)
{
```

```

    return a+b+c;
}

int main()
{
    float length,breadth;
    float base,height;
    float a,b,c;

    float (*area)(float,float);
    float (*perimeter)(float,float);

    printf("Enter Length and Breadth of Rectangle : ");
    scanf("%f%f",&length,&breadth);

    area=rectangleArea;
    perimeter=rectanglePerimeter;

    printf("Rectangle Area = %.2f\n",area(length,breadth));
    printf("Rectangle Perimeter = %.2f\n",perimeter(length,breadth));

    printf("\nEnter Base and Height of Triangle : ");
    scanf("%f%f",&base,&height);

    printf("Enter Three Sides of Triangle : ");
    scanf("%f%f%f",&a,&b,&c);

    area=triangleArea;

    printf("Triangle Area = %.2f\n",area(base,height));

    printf("Triangle Perimeter = %.2f\n",trianglePerimeter(a,b,c));

    return 0;
}

```

OUTPUT

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_3> ./10

Enter Length and Breadth of Rectangle : 10 5

Rectangle Area = 50.00

Rectangle Perimeter = 30.00

Enter Base and Height of Triangle : 6 4

Enter Three Sides of Triangle : 5 5 6

Triangle Area = 12.00

Triangle Perimeter = 16.00